



U.S. MAGNET
DEVELOPMENT
PROGRAM

Development of a 15 T Model Dipole for a Very High Energy Hadron Collider

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U.S. DEPARTMENT OF
ENERGY

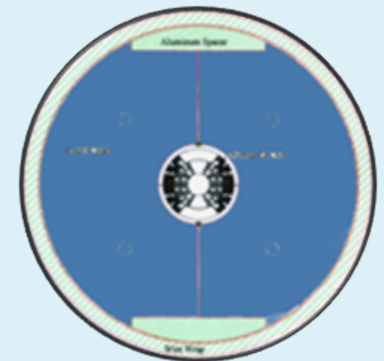
Office of
Science



Nb₃Sn magnet history

- Nb₃Sn vs. NbTi superconductor properties
 - NbTi: $T_{c0} \sim 9$ K, $B_{c20} \sim 14$ T, ductile alloy
 - Nb₃Sn: $T_{c0} \sim 18$ K, $B_{c20} \sim 28$ T, brittle compound
- NbTi accelerator magnets
 - 1970s – first IR quadrupoles in ISR at CERN
 - 1980s-2000s – Tevatron, HERA, RHIC, LHC
 - 8 T LHC dipole – the limit of the NbTi magnets
- Nb₃Sn accelerator magnets
 - 1989 – first demonstration of 10 T (CERN-ELIN)
 - 1997 – 1st attempt to achieve 15 T (D20, LBNL)
 - $B_{max} = 13.5$ T at 1.9K
 - 2008 – 2nd attempt to achieve 15 T (HD2, LBNL)
 - $B_{max} = 13.8$ T at 4.5K
 - 14 T “wall”!?

D20 (LBNL)



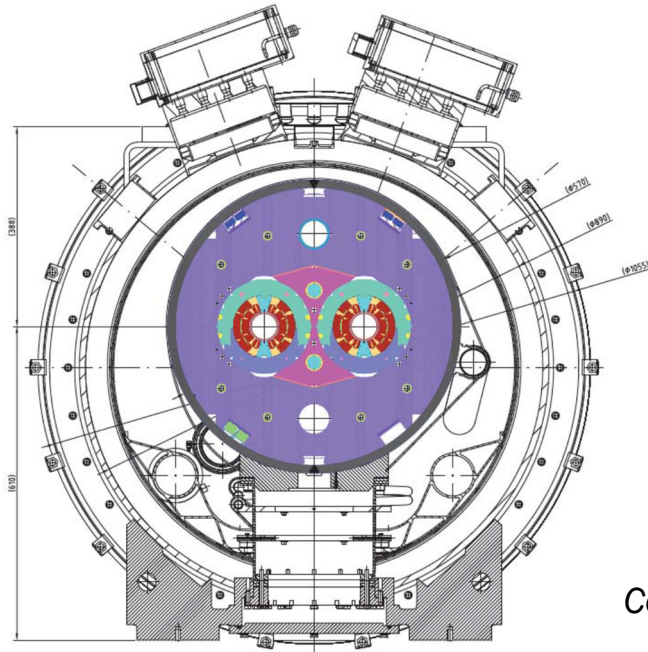
HD2 (LBNL)



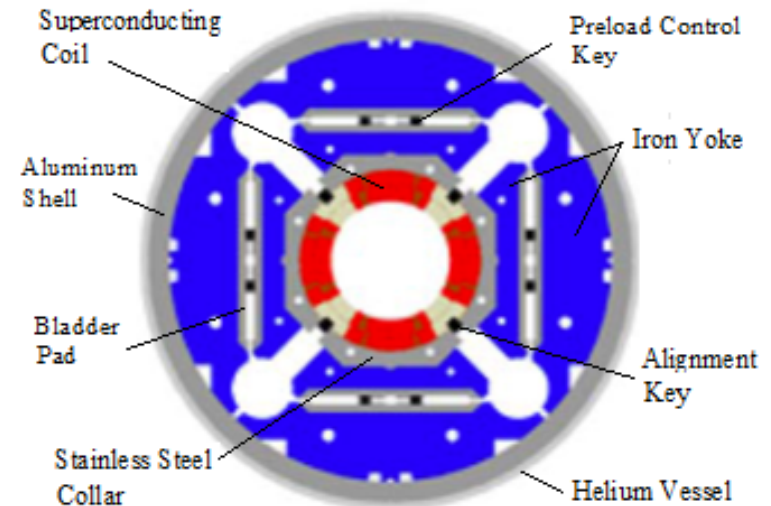


Nb₃Sn magnet progress

11 T dipole MBH for LHC
collimation system upgrade



150-mm aperture IR quadrupoles
MQXF for HL-LHC



Courtesy of CERN

- During past two decades Nb₃Sn accelerator magnet technology has made significant progress from short demonstration models to long prototypes
 - HL-LHC: 11-12 T magnets with ~20% margin



Open questions and US-MDP goals

- Open questions
 - long training
 - conductor degradation
 - maximum field limit
 - cost
- US-MDP - since 2016
- Goals and driving questions



Driving questions related to the *ultimate performance limits* of high-field accelerator magnets

- 1 What is the nature of accelerator magnet training? Can we reduce or eliminate it?
- 2 What are the drivers and required operation margin for Nb₃Sn and HTS accelerator magnets?
- 3 What are the mechanical limits and possible stress management approaches for Nb₃Sn and 20 T LTS/HTS magnets?
- 4 What are the limitations on means to safely protect Nb₃Sn and HTS magnets?

US Magnet Development Program (MDP) Goals:

GOAL 1:

Explore the performance limits of Nb₃Sn accelerator magnets with a focus on minimizing the required operating margin and significantly reducing or eliminating training.

GOAL 2:

Develop and demonstrate an HTS accelerator magnet with a self-field of 5 T or greater compatible with operation in a hybrid LTS/HTS magnet for fields beyond 16 T.

GOAL 3:

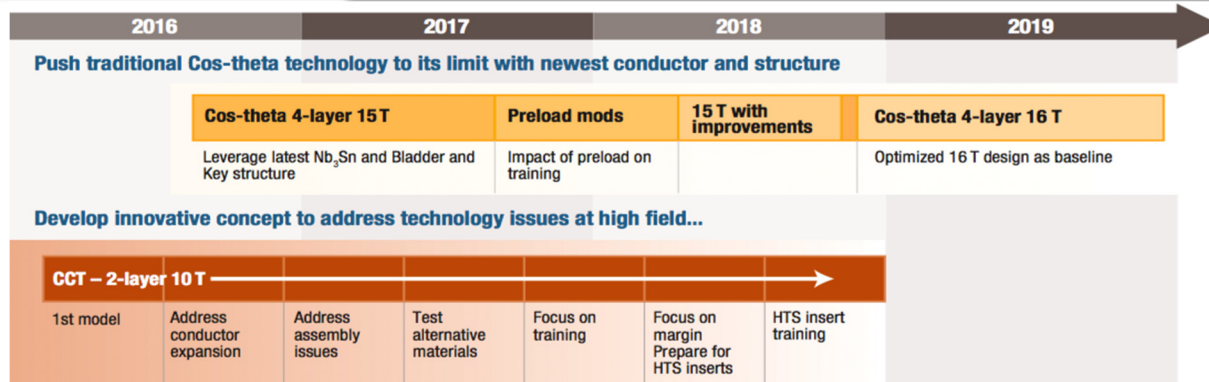
Investigate fundamental aspects of magnet design and technology that can lead to substantial performance improvements and magnet cost reduction.

GOAL 4:

Pursue Nb₃Sn and HTS conductor R&D with clear targets to increase performance and reduce the cost of accelerator magnets.



Nb₃Sn cos-theta magnet R&D plan



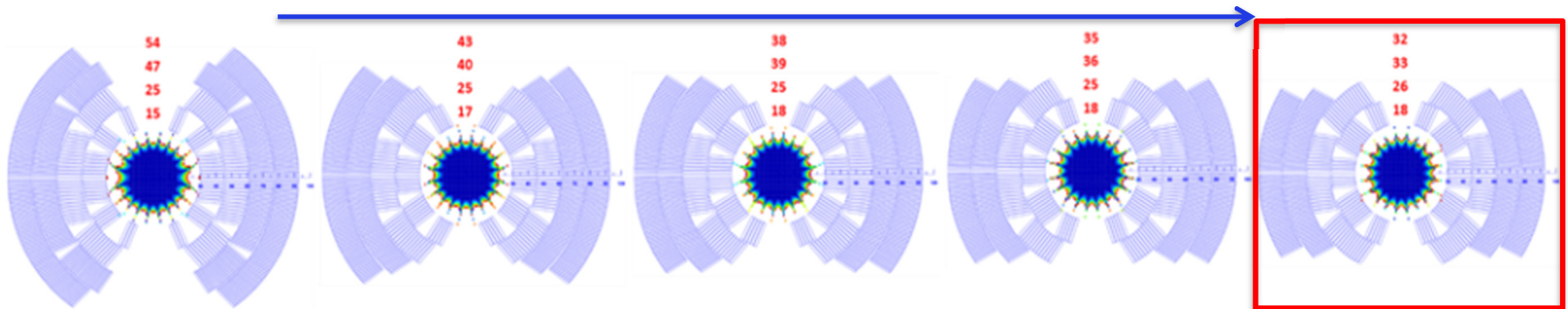
- Two design approaches: “traditional” cos-theta and innovative (canted cos-theta) dipole design concepts
- 15 T dipole demonstrator design goals:
 - explore the target field and force range
 - serve as a technical and cost basis for comparison with new concepts
 - most cost effective way to get into a field range that would exceed the LBNL D20 dipole built almost 20 years ago.
- The base for success: better design and analysis tools, new materials (Nb₃Sn wires), advanced technologies



Cos-theta coil and support structure

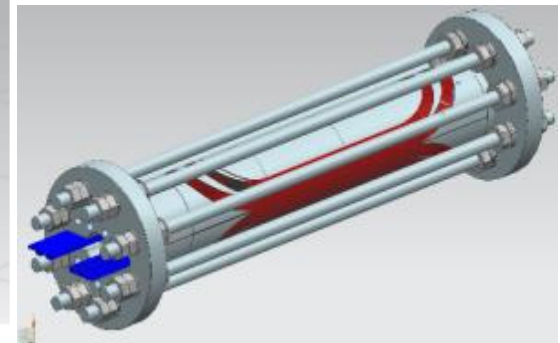
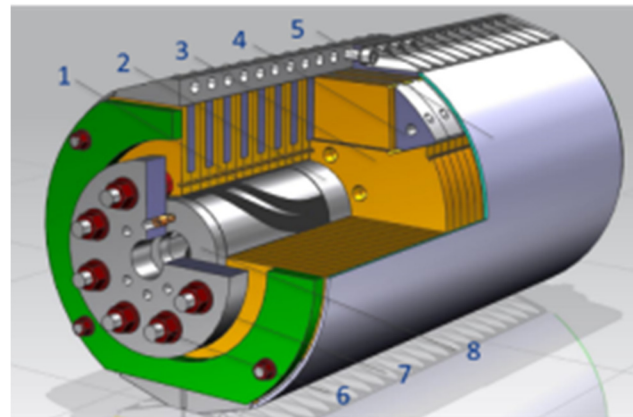
Coil: 60-mm aperture, 4-layer graded coil

Optimization criteria: field quality, azimuthal stress, protection, coil volume



Mechanical structure:

- Vertically split iron yoke
- Aluminum I-clamps
- 12-mm thick StSt skin
- StSt end plates and rods

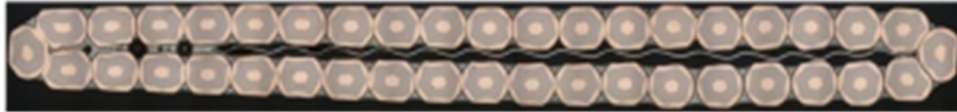




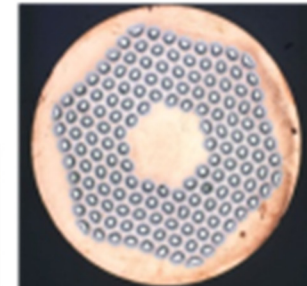
Nb₃Sn cable development



0.7 mm RRP108/127
40-strand cable with SS core



1 mm RRP150/169
28-strand cable with SS core



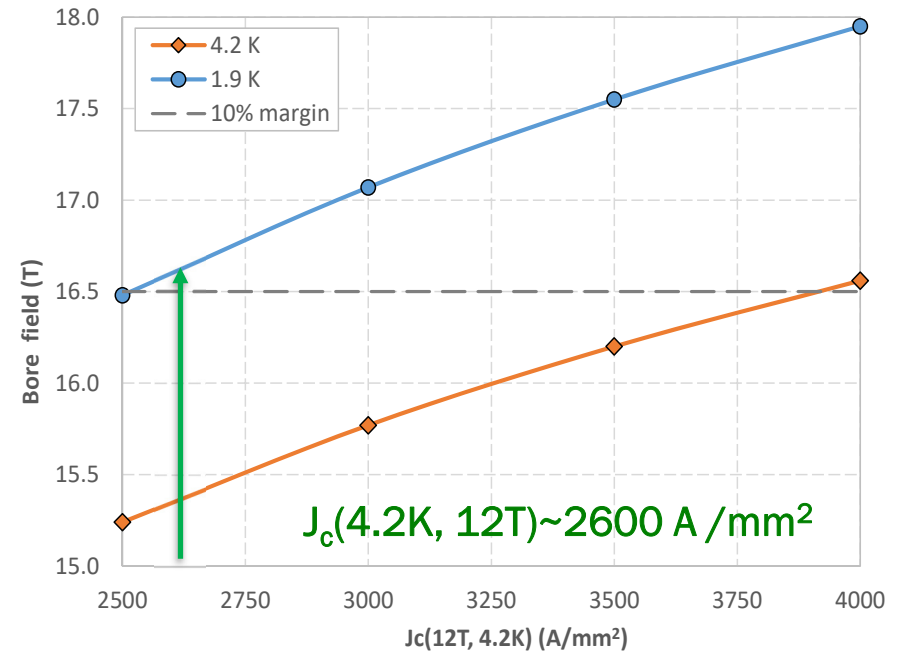
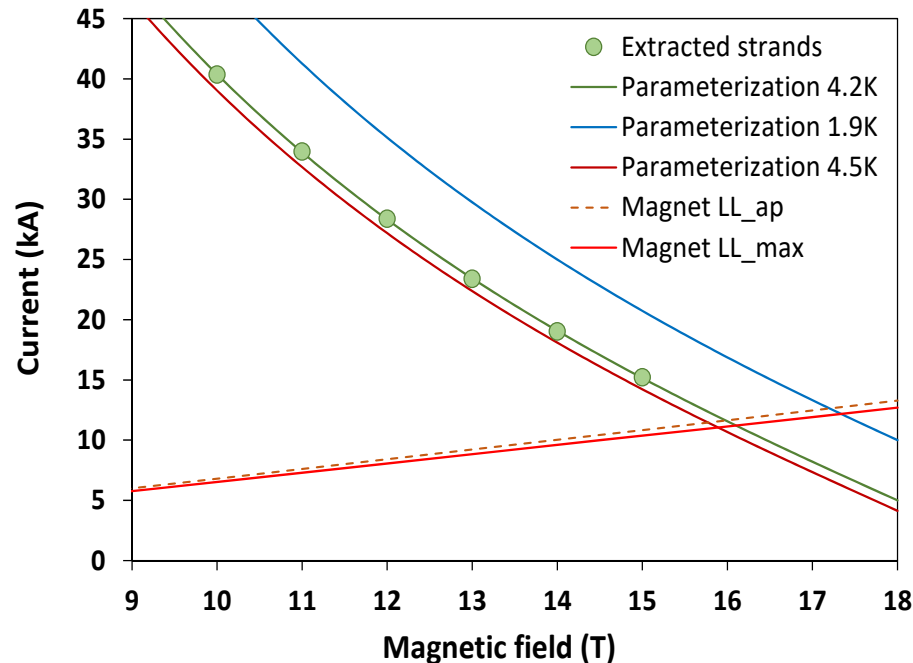
SS core

- Cable optimization
 - PF~87%,
 - I_c degradation <5%,
 - RRR>100
- HT optimization to achieve optimal J_c and RRR





Magnet conductor limit

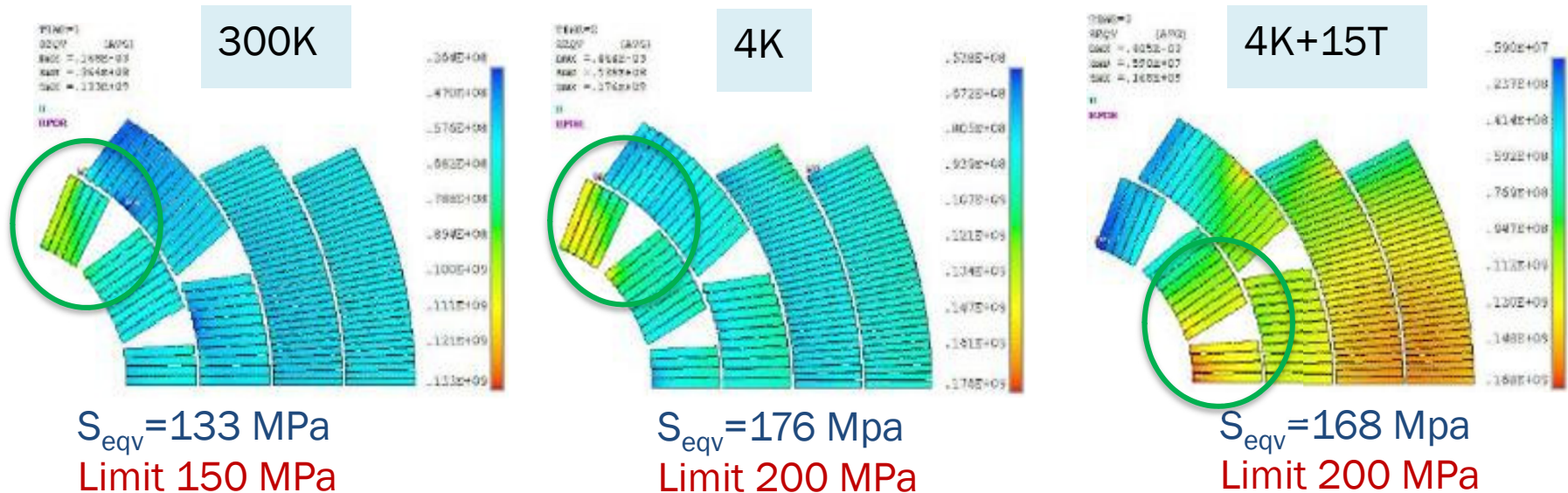


Magnet conductor limit:

- $B_{ap} = 15.3 \text{ T @ } 4.5 \text{ K}$
- $B_{ap} = 16.7 \text{ T @ } 1.9 \text{ K}$



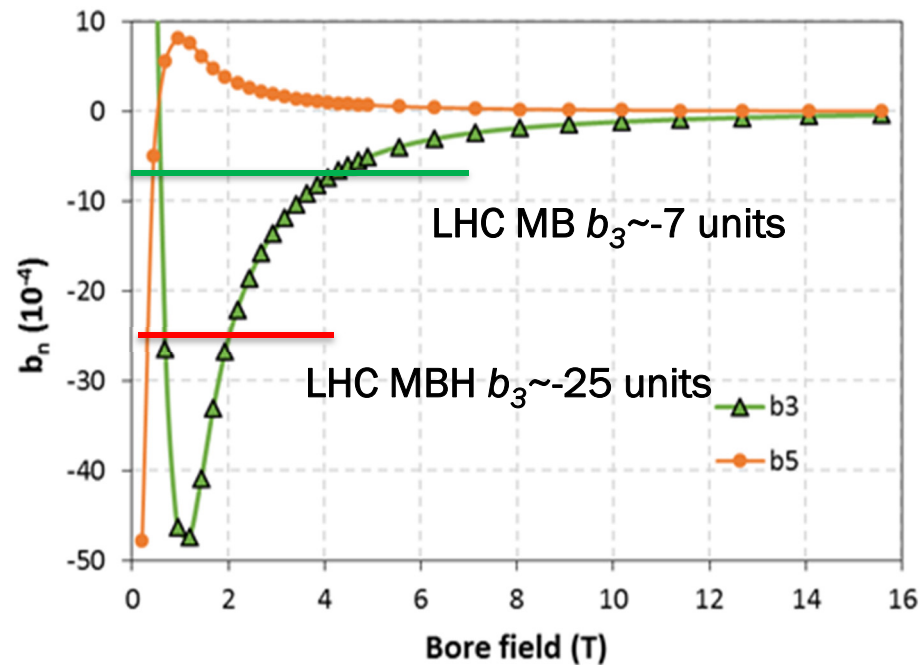
Magnet mechanical limit



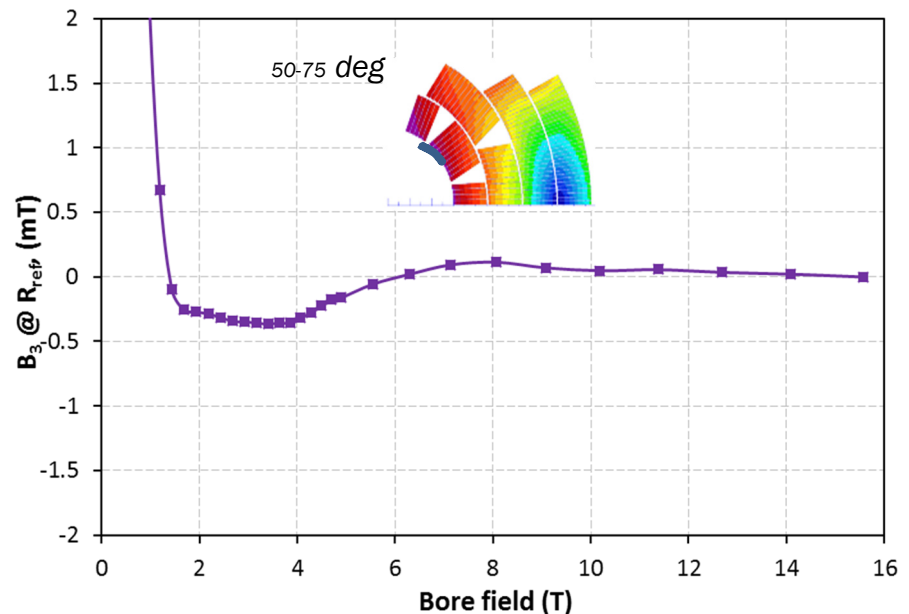
- Magnet design limit is determined by the mechanical constraints – coil maximum stress and pol turn separation from pole
- **Mechanical limit for this design is 15 T!**
- Increase of the magnet mechanical limit to 17 T is presented in poster WEPML027



Coil magnetization effect



Peak values of $b_3 = -48$ and $b_5 = +8$ units @ $B \sim 1.4$ T.



Geometrical $b_3 = -0.8$ and 0.375 mm thick iron strips



Coil components

L3/4 parts (FNAL)



Cable (FNAL)



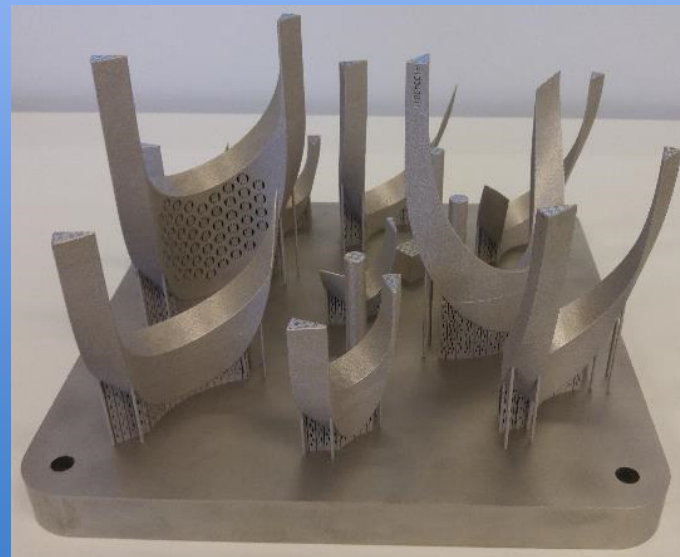
Traces (LBNL/FNAL)



L1/2 parts (CERN)



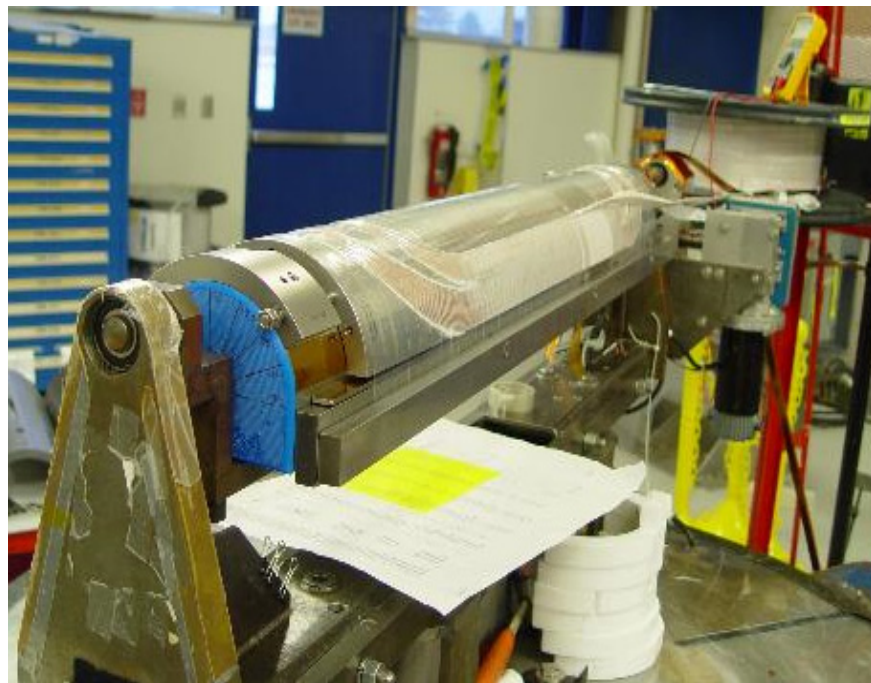
Ti and Glidcop wedges



4 sets: Ti poles and spacers, SS saddles



Coil winding and curing



- **Ceramic binder to fix turns and coil components and achieve precise coil geometry**



Coil reaction



- Coil azimuthal and radial expansion during reaction





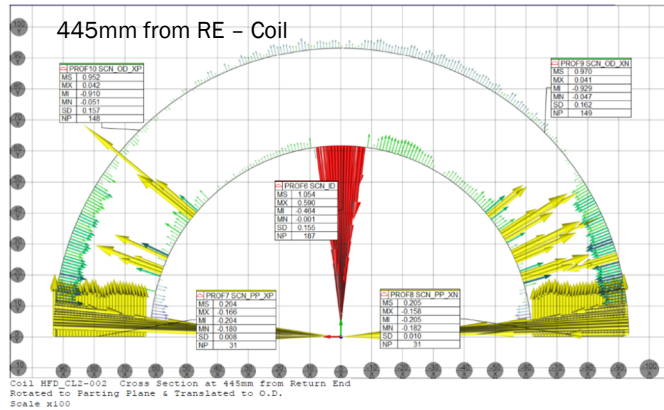
Epoxy impregnation



- Insulation mechanical and electrical reinforcement
- Eliminating coil voids (ceramic putty)



Coil size control and instrumentation



- Coil size control for coil pre-load



- Diagnostic instrumentation (VT, SG, PH, traces)



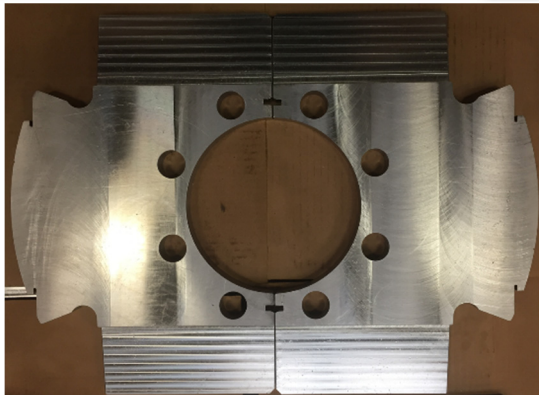
Coil fabrication status

Coil (Layer)	Coil #	Winding and curing	Reaction	Epoxy impregnatio n	Instrumenta tion
Inner (L1-L2)	1	complete	complete	in progress	
	2	complete	in progress		
	3	complete			
Outer (L3-L4)	1	complete	complete	complete	in progress
	2	complete	in progress		
	3	not started			

- Coil fabrication time ~3 months
- The goal: to have 2 IL + 2 OL coils by July



Mechanical structure components



Iron Laminations



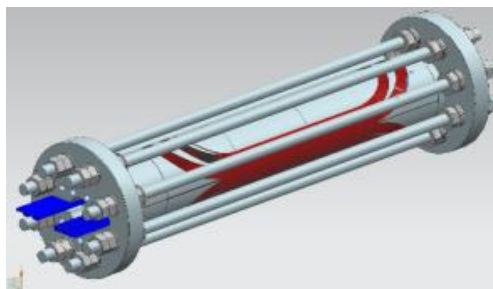
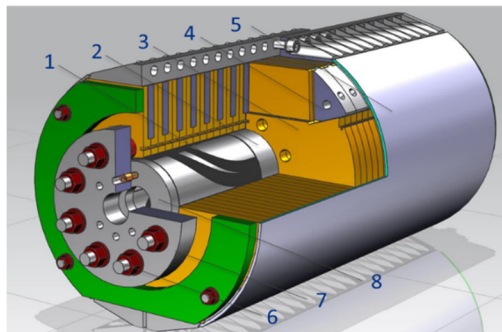
StSt Skin



AL I-Clamps



End Plates



Fillers



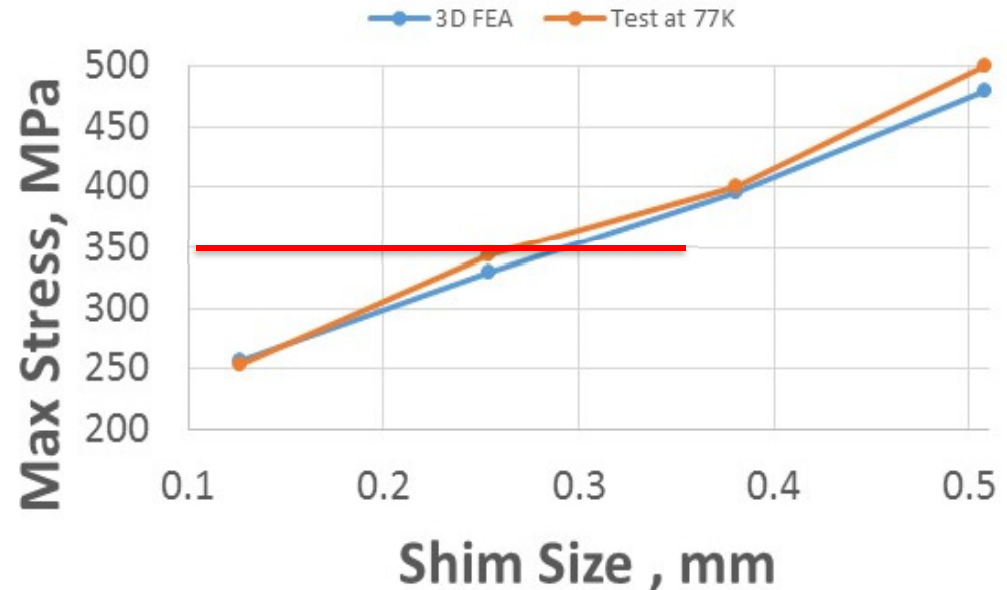
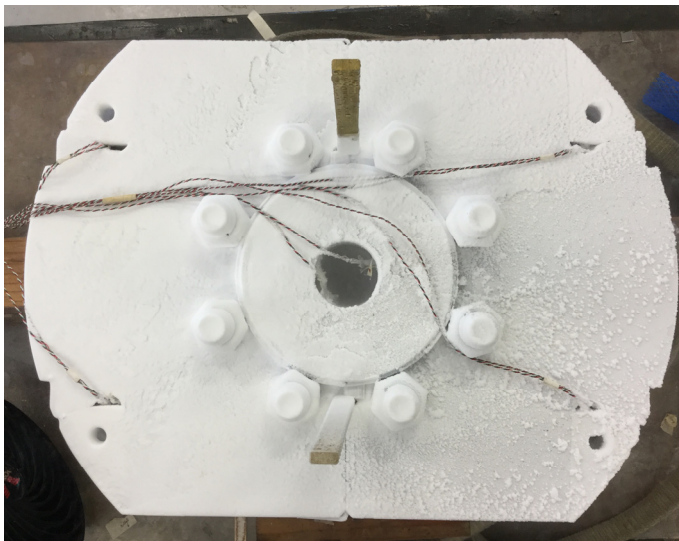
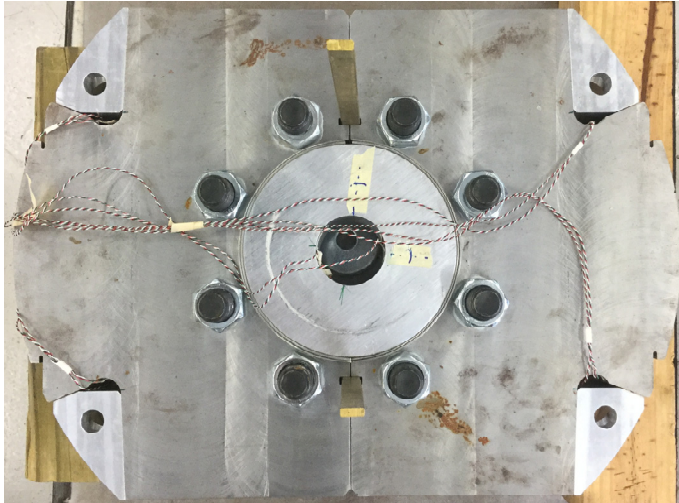
Axial Rods



Short mechanical model

Goals:

- Validate the results of mechanical analysis
- Development of the coil assembly plan and pre-stress targets
- Test all structural components, tooling and instrumentation

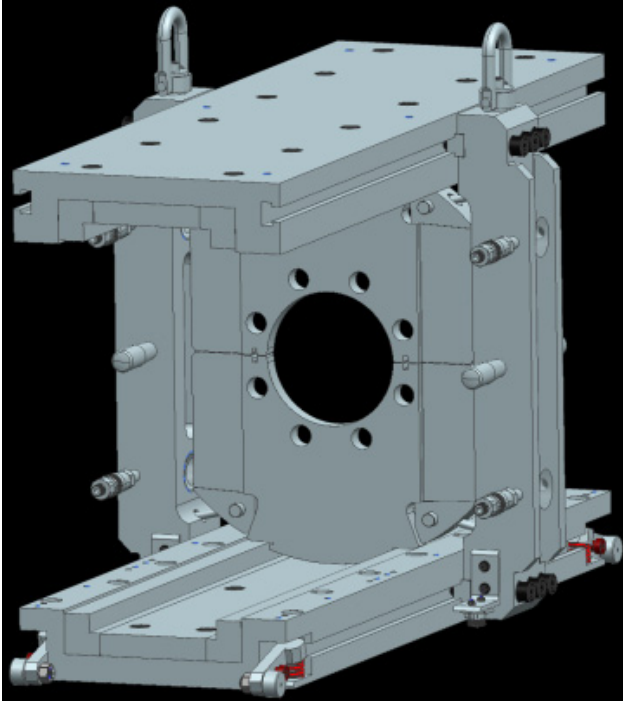
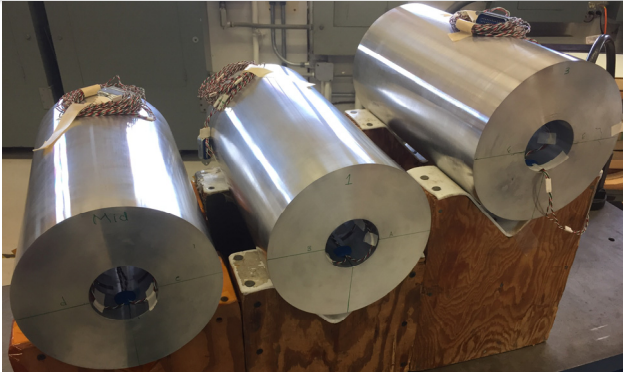




Full-scale mechanical model

Goals:

- Develop the coil assembly plan and pre-stress targets
- Test all structural components, tooling and instrumentation





Conclusions

- 15 T dipole is a challenging and important MDP milestone to understand limits of Nb₃Sn accelerator technology
- 15 T dipole demonstrator has been developed
 - optimized coil design
 - innovative mechanical structure
- Magnetic and mechanical analysis is complete
 - Conductor and mechanical limits are well understood
 - large persistent current effect in Nb₃Sn accelerator magnets was effectively reduced to the acceptable level
- Fabrication of 15 T dipole demonstrator is in progress
 - integrated international effort
- Magnet test is expected in September-October 2018



Acknowledgment

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